

Chapter 2

EXERCISE 1. Find the general solution to the following equations:

- (i) $y'(x) = e^{3x} - x$;
- (ii) $xy'(x) = 1$;
- (iii) $y'(x) = xe^{x^2}$;
- (iv) $(1+x)y'(x) = x$;
- (v) $(1+x^2)y'(x) = x$;

EXERCISE 2. Solve the following equations:

- (i) $\dot{x} = t^3(1-x)$ with $x(0) = 3$;
- (ii) $y' = (1+y^2)\tan x$ with $y(0) = 1$;
- (iii) $\dot{x} = t^2x$ (general solution);
- (iv) $\dot{x} = -x^2$ (general solution);
- (v) $dy/dt = e^{-t^2}y^2$: give the solution in terms of an integral and describe the behavior as $t \rightarrow \infty$ depending on the initial condition $y(0)$. Use that $\int_0^\infty e^{-s^2} ds = \sqrt{\pi}/2$.

EXERCISE 3. Use an integrating factor to solve the following differential equations:

- (i) $\frac{dy}{dx} + \frac{y}{x} = x^2$ (find the general solution and the only solution that is finite when $x = 0$),
- (ii) $\frac{dx}{dt} + tx = 4t$ (find the solution with $x(0) = 2$),
- (iii) $\frac{dz}{dy} = z \tan y + \sin y$ (find the general solution),
- (iv) $y' + e^{-x}y = 1$ (find the solution when $y(0) = e$, leaving your answer as an integral),
- (v) $\dot{x} + x \tanh t = 3$ (find the general solution, and compare it to that for $\dot{x} + x = 3$),
- (vi) $y' + 2y \cot x = 5$ (find the solution with $y(\pi/2) = 1$),
- (vii) $\frac{dx}{dt} + 5x = t$ (find the general solution),
- (viii) with $a > 0$, find the solution of the equation $\frac{dx}{dt} + \left[a + \frac{1}{t}\right]x = b$ for a general initial condition $x(1) = x_0$, and show that $x(t) \rightarrow b/a$ as $t \rightarrow \infty$.

EXERCISE 4. Check that the following equations are exact and solve them.

(i) $(2xy - \sec^2 x) + (x^2 + 2y) \frac{dy}{dx} = 0,$

(ii) $(1 + e^x y + x e^x y) + (x e^x + 2) \frac{dy}{dx} = 0,$

(iii) $(x \cos y + \cos x) \frac{dy}{dx} + \sin y - y \sin x = 0,$

(iv) $e^x \sin y + y + (e^x \cos y + x + e^y) \frac{dy}{dx} = 0.$

EXERCISE 5. Find an integrating factor depending only on x that makes the equation

$$3xy + y^2 + (x^2 + xy) \frac{dy}{dx} = 0$$

exact, and find its solution.

EXERCISE 6. Show that any equation that can be written in the form

$$f(x) + g(y) \frac{dy}{dx} = 0$$

is exact, and find its solution in terms of integrals of f and g . Hence find the solutions of

(i) $V'(x) + 2y \frac{dy}{dx} = 0,$

(ii) $\left[\frac{1}{y} - a \right] \frac{dy}{dx} + \frac{2}{x} - b = 0, \text{ for } x, y > 0.$

EXERCISE 7. We consider the following initial value problem:

$$y'(x) = y(x) + \frac{x}{y(x)}, \quad y(0) = 1.$$

Make the change of variable $z(x) = y(x)^2$ and find the solution.

EXERCISE 8. Show that the Bernoulli equation

$$y'(x) = a(x)y(x) + b(x)y(x)^\alpha$$

where $\alpha \neq 0, 1$, can be transformed into a linear ODE with the change of variable $z(x) = y(x)^{1-\alpha}$ and find the solution. What happens if $\alpha = 0$ or $\alpha = 1$?

EXERCISE 9. Which of the following differential equations have unique solutions (at least on some small time interval) for any non-negative initial condition $x(0) \geq 0$?

(i) $\dot{x} = x(1 - x^2),$

- (ii) $\dot{x} = x^3$,
- (iii) $\dot{x} = x^{1/3}$,
- (iv) $\dot{x} = x^{1/2}(1+x)^2$,
- (v) $\dot{x} = (1+x)^{3/2}$.

EXERCISE 10. Without using Picard's theorem, show that the IVP

$$\frac{dy}{dt} + p(t)y = 0, \quad y(t_0) = y_0$$

with $p(t)$ a continuous function has a unique solution, and calculate it. What solution do we obtain if $y_0 = 0$? Hint: consider the function $z(t) = y(t)e^{\int_{t_0}^t p(s)ds}$, where $y(t)$ solves the IVP, show that $z(t)$ is constant and deduce $y(t)$.

EXERCISE 11. Without solving the ODE, show that if $y_1(t)$ and $y_2(t)$ are any two solutions of

$$\frac{dy}{dt} + p(t)y = 0,$$

then the ratio $y_1(t)/y_2(t)$ is constant.

EXERCISE 12. Consider $\frac{dy}{dt} = f(y, t)$, where f satisfies the hypothesis of the existence and uniqueness theorem. Suppose that

- (i) $y_1(t) = -2$ for all t is a solution and we are studying a different solution for which $y(0) = 0$.
- (ii) $y_1(t) = -t - 1$ and $y_2(t) = t^2 + 1$ are solutions and $y(0) = 0$.

Based on the uniqueness theorem, what can you conclude about the solutions in each case?

EXERCISE 13. Consider the autonomous differential equation

$$\frac{dy}{dt} = y(y-2)(y-3).$$

In each of the following cases, what can you conclude based on the uniqueness theorem about the different solutions if

- (i) $y(0) = -1$,
- (ii) $y(0) = 1$,
- (iii) $y(0) = 2$,
- (iv) $y(0) = 4$?

EXERCISE 14. Given the IVP

$$\frac{dy}{dt} = 3y^{2/3}, \quad y(0) = 0,$$

on the rectangle $R = \{(y, t) \in \mathbb{R}^2 : |y|, |t| \leq 1\}$. Show that both $y_1(t) = t^3$ and $y_2(t) = 0$ are solutions. Does this contradict Picard's theorem?

EXERCISE 15. Consider the differential equation

$$\frac{dy}{dt} = \frac{2y+1}{t}.$$

Compute the general solution and show that there are two solutions such that $y(0) = -1/2$. Does this contradict the uniqueness theorem?

EXERCISE 16. Consider the autonomous equation

$$\frac{dx}{dt} = x(\kappa^2 - x^2)$$

with a general initial condition $x(0) = x_0$.

- (i) Calculate the stationary solutions, which satisfy $\frac{dx}{dt} = 0$;
 - (ii) Calculate and sketch the solutions for $t > 0$, discussing the different behaviors which can be obtained in terms of the relative values of $\kappa > 0$ and x_0 , with the help of the uniqueness theorem.
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EXERCISE 17. Show existence and uniqueness for the initial value problem (IVP)

$$\frac{dy}{dt} + ay = 0, \quad y(t_0) = y_0.$$

For existence, just find a solution by inspection. Then suppose that there are two different solutions and argue that they have to be the same by taking their difference.
